A Verified Partial Convexification for Finite Vertex-Listed sa-Rectangular Robust Markov Decision Processes

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Abstract

This note records a restricted positive result for the convex-formulation problem for robust Markov decision processes. The result is *partial progress*, not a solution of the general problem. For a discounted robust MDP with rational data and finite, explicitly vertex-listed, state-action rectangular transition ambiguity, a single convex lifted program gives a point closer to the exact robust value vector than the rational separation scale of all possible exact values. Continued-fraction reconstruction then recovers the exact value vector and hence an optimal stationary deterministic robust policy. The construction is best understood as an exact rounding version of the exponential/geometric change of variables used in convex formulations of robust MDPs. It does not cover general ambiguity sets such as ellipsoidal, Wasserstein, or divergence balls unless they are supplied by a polynomial vertex list, and it should not be interpreted as a numerically practical polynomial-time algorithm.

Status relative to the Grand-Clement–Petrik problem. The theorem below gives an exact, succinct convex formulation only for rational discounted RMDPs with polynomially listed finite-vertex sa-rectangular ambiguity. Thus it is meaningful partial progress on the convex-formulation issue, but it is not a solution of the authors' full problem for general robust MDP ambiguity models.

1 Model and normalization

Let $S = \{1, \dots, n\}$ be a finite state space. For each state s, let A_s be a nonempty finite action set. For each state-action pair (s, a), the one-step reward is rational and denoted by r_{sa}. The discount factor is a rational number $\gamma \in (0, 1)$. The transition ambiguity set is assumed to be state-action rectangular and given as the convex hull of a finite, explicitly listed set of rational probability vectors,

$$\mathcal{U}_{sa} = \operatorname{conv}\{p_{sa}^b : b \in B_{sa}\} \subseteq \Delta(S). \tag{1.1}$$

Because the continuation term is linear in the transition vector, for every $x \in \mathbb{R}^S$,

$$\min_{p \in \mathcal{U}_{sa}} p^\top x = \min_{b \in B_{sa}} (p_{sa}^b)^\top x. \tag{1.2}$$

The robust Bellman operator is

$$(Tv)_s = \max_{a \in A_s} \left\{ r_{sa} + \gamma \min_{b \in B_{sa}} (p_{sa}^b)^\top v \right\}. \tag{1.3}$$

It is monotone and is a γ-contraction in $\|\cdot\|_\infty$. Hence it has a unique fixed point $v^\star$, the robust optimal value vector.

We reduce to nonnegative rewards in a unit interval. Let

$$r_{\min} := \min_{s,a} r_{sa}, \qquad r_{\max} := \max_{s,a} r_{sa}, \qquad R := r_{\max} - r_{\min}. \tag{1.4}$$

If $R = 0$, then every action gives the same one-step reward $r_{\min}$, so $v^\star = (r_{\min}/(1-\gamma))\mathbf{1}$, and the problem is trivial. Assume from now on that $R > 0$, and define shifted rewards

$$\bar{r}_{sa} := r_{sa} - r_{\min} \in [0, R]. \tag{1.5}$$

Let $\bar{T}$ be the Bellman operator obtained from (1.3) by replacing r_{sa} by $\bar{r}_{sa}$, and let $\bar{v}^\star$ be its fixed point. Then

$$v^\star = \bar{v}^\star + \frac{r_{\min}}{1-\gamma}\mathbf{1}. \tag{1.6}$$

Thus it suffices to recover $\bar{v}^\star$. Define the scaled value

$$u^\star := \frac{1-\gamma}{R}\bar{v}^\star \in [0,1]^S \tag{1.7}$$

and the scaled rewards

$$c_{sa} := \frac{(1-\gamma)\bar{r}_{sa}}{R} \in [0, 1-\gamma]. \tag{1.8}$$

Then $u^\star$ is the unique fixed point of

$$F_s(u) = \max_{a \in A_s} \left\{ c_{sa} + \gamma \min_{b \in B_{sa}} (p^b_{sa})^\top u \right\}. \tag{1.9}$$

The operator F is monotone, is a γ-contraction in $\|\cdot\|_\infty$, and satisfies the translation identity

$$F(u + \alpha\mathbf{1}) = F(u) + \gamma\alpha\mathbf{1} \qquad (\alpha \in \mathbb{R}). \tag{1.10}$$

2 The lifted convex program

Let

$$A_{\max} := \max_{s \in S} |A_s|. \tag{2.1}$$

Let Q be a positive integer such that every number c_{sa} and every number $\gamma p^b_{sa}(t)$ has denominator dividing Q. Set

$$H := \left\lceil (\sqrt{n}\, Q)^n \right\rceil. \tag{2.2}$$

Choose a base $M > 1$ satisfying

$$\frac{n \log_M A_{\max}}{1-\gamma} < \frac{1}{4H^2}, \tag{2.3}$$

where $\log_M 1 = 0$. The large size of M is important; see Section 8 for the precise interpretation of this choice.

Introduce two terminal nodes W, L, and for each state-action pair (s,a), introduce two auxiliary nodes C_{sa} and D_{sa}. The variables are positive real numbers

$$Y_i > 0, \qquad i \in S \cup \{W, L\} \cup \{C_{sa} : s \in S, a \in A_s\} \cup \{D_{sa} : s \in S, a \in A_s\}. \tag{2.4}$$

For each (s,a), define

$$\rho_{sa} := c_{sa}, \qquad \lambda_{sa} := (1-\gamma) - c_{sa}. \tag{2.5}$$

Then $\rho_{sa}, \lambda_{sa}, \gamma \geq 0$ and $\rho_{sa} + \lambda_{sa} + \gamma = 1$. Consider the convex optimization problem

$$\begin{array}{lll} \text{maximize} & \displaystyle\sum_{s\in S} \log Y_s & \\ \text{subject to} & 1 \leq Y_i \leq M & \text{for every nonterminal node } i, \\ & Y_W = M, \qquad Y_L = 1, & \\ & Y_s \leq \displaystyle\sum_{a\in A_s} Y_{C_{sa}} & \text{for every } s \in S, \\ & Y_{C_{sa}} \leq Y_W^{\rho_{sa}} Y_L^{\lambda_{sa}} Y_{D_{sa}}^{\gamma} & \text{for every } (s, a), \\ & Y_{D_{sa}} \leq \displaystyle\prod_{t\in S} Y_t^{p_{sa}^b(t)} & \text{for every } (s, a) \text{ and } b \in B_{sa}. \end{array} \tag{2.6}$$

Here "nonterminal" means every variable except Y_W and Y_L. The terminal equalities already imply $Y_W, Y_L \in [1, M]$.

Proposition 2.1 (Convexity)**.** *Problem* (2.6) *is a convex optimization problem. More precisely, the feasible region is convex, and the objective is concave.*

Proof. The constraints

$$Y_{C_{sa}} \leq Y_W^{\rho_{sa}} Y_L^{\lambda_{sa}} Y_{D_{sa}}^{\gamma}, \qquad Y_{D_{sa}} \leq \prod_{t\in S} Y_t^{p_{sa}^b(t)} \tag{2.7}$$

are hypographs of weighted geometric means on the positive orthant. Weighted geometric means are concave, so these hypographs are convex. The remaining constraints are linear inequalities or linear equalities. Finally, $Y \mapsto \sum_s \log Y_s$ is concave on the positive orthant. Therefore maximizing this objective over the displayed feasible region is a convex optimization problem in the standard sense. ◻

Remark 2.2 (Conic representation)**.** *For rational weights, the weighted geometric-mean hypograph constraints can be represented by weighted geometric-mean cones, equivalently by power cones. If one wants a purely conic form for the logarithmic objective, introduce variables z_s with $z_s \leq \log Y_s$, equivalently $\exp(z_s) \leq Y_s$, and maximize $\sum_s z_s$ using exponential cones. None of the exactness argument depends on this modeling choice.*

3 Main theorem

Theorem 3.1 (Exact recovery for the finite-vertex case)**.** *For every rational discounted RMDP satisfying the finite vertex-listed sa-rectangular assumptions above, choose M satisfying* (2.3)*. Let $\widehat{Y}$ be any optimizer of* (2.6)*, and define*

$$\widehat{u}_s := \log_M \widehat{Y}_s, \qquad s \in S. \tag{3.1}$$

Then

$$\|\widehat{u} - u^\star\|_\infty < \frac{1}{4H^2}. \tag{3.2}$$

Consequently, each coordinate $u_s^\star$ is the unique rational number of denominator at most H lying in the interval

$$\left(\widehat{u}_s - \frac{1}{4H^2}, \widehat{u}_s + \frac{1}{4H^2}\right). \tag{3.3}$$

Thus $u^\star$ is recovered exactly by rational reconstruction. The original robust value is then

$$v_s^\star = \frac{R}{1-\gamma} u_s^\star + \frac{r_{\min}}{1-\gamma}, \qquad s \in S. \tag{3.4}$$

Finally, any stationary deterministic policy $\pi^\star$ satisfying

$$\pi^\star(s) \in \arg\max_{a \in A_s} \left\{ r_{sa} + \gamma \min_{b \in B_{sa}} (p_{sa}^b)^\top v^\star \right\} \tag{3.5}$$

for every state s is robust-optimal.

The rest of the note proves the theorem.

4 Rational separation

Lemma 4.1 (Denominator bound)**.** *Every coordinate of $u^\star$ is rational and has denominator at most H, with H defined by* (2.2)*.*

Proof. For every state s, choose an action $a_s \in A_s$ attaining the maximum in (1.9) at $u^\star$. For every chosen pair (s, a_s), choose a vertex $b_s \in B_{sa_s}$ attaining the corresponding minimum at $u^\star$. These choices are possible because all sets involved are finite. Then $u^\star$ satisfies the linear system

$$u_s^\star = c_{sa_s} + \gamma \sum_{t \in S} p_{sa_s}^{b_s}(t) u_t^\star, \qquad s = 1, \dots, n. \tag{4.1}$$

Let P be the stochastic matrix whose s-th row is $p_{sa_s}^{b_s}$, and let $c \in \mathbb{Q}^n$ have coordinates c_{sa_s}. Then

$$(I - \gamma P) u^\star = c. \tag{4.2}$$

Since $\rho(\gamma P) \le \gamma < 1$, the matrix $I - \gamma P$ is invertible.

By the definition of Q, the matrix

$$A := Q(I - \gamma P) \tag{4.3}$$

has integer entries, and the vector $b := Qc$ has integer entries. The system is equivalently

$$A u^\star = b. \tag{4.4}$$

Each entry of A has absolute value at most Q, so each row of A has Euclidean norm at most $\sqrt{n}\, Q$. Hadamard's inequality gives

$$|\det A| \le (\sqrt{n}\, Q)^n \le H. \tag{4.5}$$

By Cramer's rule, each coordinate of $u^\star = A^{-1} b$ is rational with denominator dividing $|\det A|$, after reduction. Hence every denominator is at most H. □

Corollary 4.2 (Rational separation)**.** *If $x \ne y$ are rational numbers whose denominators are both at most H, then*

$$|x - y| \ge \frac{1}{H^2}. \tag{4.6}$$

In particular, an interval of radius $1/(4H^2)$ contains at most one rational number with denominator at most H.

Proof. Write $x = a/b$ and $y = c/d$ in lowest terms with $1 \le b, d \le H$. Then

$$|x - y| = \frac{|ad - bc|}{bd} \ge \frac{1}{bd} \ge \frac{1}{H^2}, \tag{4.7}$$

provided $x \ne y$. The interval statement follows immediately. □

5 Approximate subsolutions from the convex lift

Lemma 5.1 (Every feasible lift gives an approximate Bellman subsolution)**.** *Let Y be feasible for* (2.6)*, and define $u_s := \log_M Y_s$. Then $u \in [0,1]^S$ and*

$$u \le F(u) + \delta \mathbf{1}, \qquad \delta := \log_M A_{\max}. \tag{5.1}$$

Proof. The bounds $1 \le Y_s \le M$ imply $u_s \in [0,1]$. For every (s,a) and $b \in B_{sa}$, the constraint on D_{sa} gives

$$\log_M Y_{D_{sa}} \le \sum_{t \in S} p_{sa}^b(t) \log_M Y_t = (p_{sa}^b)^\top u. \tag{5.2}$$

Because this holds for every b,

$$\log_M Y_{D_{sa}} \le \min_{b \in B_{sa}} (p_{sa}^b)^\top u. \tag{5.3}$$

The constraint on C_{sa}, together with $Y_W = M$, $Y_L = 1$, and $\rho_{sa} = c_{sa}$, gives

$$\begin{aligned}\log_M Y_{C_{sa}} &\le \rho_{sa} \log_M Y_W + \lambda_{sa} \log_M Y_L + \gamma \log_M Y_{D_{sa}} \\ &= c_{sa} + \gamma \log_M Y_{D_{sa}}.\end{aligned} \tag{5.4}$$

Combining (5.3) and (5.4),

$$\log_M Y_{C_{sa}} \le c_{sa} + \gamma \min_{b \in B_{sa}} (p_{sa}^b)^\top u. \tag{5.5}$$

Finally, the state constraint gives

$$\begin{aligned}u_s = \log_M Y_s &\le \log_M \left(\sum_{a \in A_s} Y_{C_{sa}} \right) \\ &\le \max_{a \in A_s} \log_M Y_{C_{sa}} + \log_M |A_s| \\ &\le F_s(u) + \log_M A_{\max}.\end{aligned} \tag{5.6}$$

This is (5.1). ☐

Lemma 5.2 (Approximate subsolutions lie near the fixed point)**.** *If $u \in [0,1]^S$ satisfies $u \le F(u) + \delta \mathbf{1}$, then*

$$u \le u^\star + \frac{\delta}{1-\gamma} \mathbf{1}. \tag{5.7}$$

Proof. Using monotonicity and the translation identity (1.10), we prove by induction that for every $k \ge 1$,

$$u \le F^k(u) + \delta(1 + \gamma + \cdots + \gamma^{k-1}) \mathbf{1}. \tag{5.8}$$

The case $k = 1$ is the hypothesis. If (5.8) holds for k, then

$$\begin{aligned} u &\le F^k(u) + \delta(1 + \cdots + \gamma^{k-1}) \mathbf{1}, \\ F(u) &\le F^{k+1}(u) + \gamma\delta(1 + \cdots + \gamma^{k-1}) \mathbf{1}. \end{aligned}$$

The second inequality is obtained by applying F to the induction bound. Combining it with $u \le F(u) + \delta \mathbf{1}$ yields the case $k+1$. Since F is a contraction, $F^k(u) \to u^\star$ as $k \to \infty$. Letting $k \to \infty$ in (5.8) gives (5.7). ☐

6 Optimality gives the missing lower bound

The previous lemma gives only an upper coordinatewise bound on a lifted feasible point. The convex objective supplies the complementary aggregate lower bound.

Lemma 6.1 (The exact value lift is feasible)**.** *There is a feasible point $Y^\star$ of* (2.6) *satisfying*

$$Y_s^\star = M^{u_s^\star}, \qquad s \in S. \tag{6.1}$$

Proof. Set $Y_W^\star = M$, $Y_L^\star = 1$, and $Y_s^\star = M^{u_s^\star}$. For every (s, a), set

$$d_{sa} := \min_{b \in B_{sa}} (p_{sa}^b)^\top u^\star, \qquad Y_{D_{sa}}^\star := M^{d_{sa}}, \tag{6.2}$$

and

$$Y_{C_{sa}}^\star := M^{c_{sa} + \gamma d_{sa}}. \tag{6.3}$$

Since $u^\star \in [0,1]^S$, we have $d_{sa} \in [0,1]$. Since $c_{sa} \in [0, 1-\gamma]$, also $c_{sa} + \gamma d_{sa} \in [0,1]$. Hence all nonterminal variables lie in $[1, M]$.

For every $b \in B_{sa}$,

$$Y_{D_{sa}}^\star = M^{d_{sa}} \le M^{(p_{sa}^b)^\top u^\star} = \prod_t (Y_t^\star)^{p_{sa}^b(t)}. \tag{6.4}$$

Also,

$$(Y_W^\star)^{\rho_{sa}} (Y_L^\star)^{\lambda_{sa}} (Y_{D_{sa}}^\star)^\gamma = M^{\rho_{sa}} \cdot 1^{\lambda_{sa}} \cdot M^{\gamma d_{sa}} = M^{c_{sa} + \gamma d_{sa}} = Y_{C_{sa}}^\star. \tag{6.5}$$

Finally, because $u^\star = F(u^\star)$,

$$M^{u_s^\star} = \max_{a \in A_s} M^{c_{sa} + \gamma d_{sa}} = \max_{a \in A_s} Y_{C_{sa}}^\star \le \sum_{a \in A_s} Y_{C_{sa}}^\star. \tag{6.6}$$

Thus all constraints are satisfied. □

Lemma 6.2 (Optimizer closeness)**.** *Let $\widehat{Y}$ be an optimizer of* (2.6) *and set $\widehat{u}_s = \log_M \widehat{Y}_s$. Then*

$$\|\widehat{u} - u^\star\|_\infty \le \frac{n \log_M A_{\max}}{1 - \gamma}. \tag{6.7}$$

Proof. Let $\delta := \log_M A_{\max}$ and $\Delta := \delta/(1-\gamma)$. By Lemmas 5.1 and 5.2,

$$\widehat{u}_s \le u_s^\star + \Delta \qquad \text{for all } s. \tag{6.8}$$

By Lemma 6.1, the exact lift $Y^\star$ is feasible. Since $\widehat{Y}$ maximizes $\sum_s \log Y_s$,

$$\sum_s \log \widehat{Y}_s \ge \sum_s \log Y_s^\star. \tag{6.9}$$

Dividing by $\log M$,

$$\sum_s \widehat{u}_s \ge \sum_s u_s^\star. \tag{6.10}$$

If some coordinate k satisfied $\widehat{u}_k < u_k^\star - n\Delta$, then using (6.8) on all other coordinates would give

$$\begin{aligned} \sum_s \widehat{u}_s &< u_k^\star - n\Delta + \sum_{s \ne k} (u_s^\star + \Delta) \\ &= \sum_s u_s^\star - \Delta, \end{aligned}$$

contradicting (6.10). Hence

$$\widehat{u}_s \ge u_s^\star - n\Delta \qquad \text{for all } s. \tag{6.11}$$

Together with (6.8), and using $n \ge 1$, this proves (6.7). □

7 Proof of the main theorem

Proof of Theorem 3.1. By Lemma 6.2 and the choice of M,

$$\|\widehat{u} - u^\star\|_\infty \le \frac{n \log_M A_{\max}}{1-\gamma} < \frac{1}{4H^2}. \tag{7.1}$$

This proves (3.2). By Lemma 4.1, each coordinate of $u^\star$ has denominator at most H. By Corollary 4.2, no interval of radius $1/(4H^2)$ can contain two distinct rationals with denominator at most H. Therefore the interval around $\widehat{u}_s$ contains exactly one such rational, namely $u_s^\star$. Continued fractions, or any standard rational reconstruction algorithm with denominator bound H, recovers $u_s^\star$ exactly.

Equation (3.4) follows from the scaling relation (1.7) and the reward-shift relation (1.6). Finally, let $\pi^\star$ satisfy (3.5). Define its robust policy-evaluation operator

$$(T^{\pi^\star} v)_s = r_{s\pi^\star(s)} + \gamma \min_{b \in B_{s\pi^\star(s)}} (p^b_{s\pi^\star(s)})^\top v. \tag{7.2}$$

The operator $T^{\pi^\star}$ is a γ-contraction, and by greediness,

$$T^{\pi^\star} v^\star = T v^\star = v^\star. \tag{7.3}$$

Thus $v^\star$ is the robust value of $\pi^\star$. Since $v^\star$ is the optimal robust fixed point of T, the policy $\pi^\star$ is robust-optimal. □

8 Size, interpretation, and limitations

The number of ordinary variables in (2.6) is

$$n + 2 + 2\sum_{s \in S} |A_s|. \tag{8.1}$$

The number of state max constraints is n, the number of reward/continuation geometric-mean constraints is $\sum_s |A_s|$, and the number of adversarial geometric-mean constraints is $\sum_{s,a} |B_{sa}|$. Thus the visible lifted model is polynomial in the explicit vertex-list input.

The subtlety is the scalar M. The denominator bound has

$$\log H = O\big(n(\log Q + \log n)\big), \tag{8.2}$$

so condition (2.3) can be met with $M = 2^{2^K}$, where K is polynomially bounded in the bit length of the rational input. Written literally, however, M has exponentially many bits. Hence a literal coefficient encoding would destroy any polynomial-size claim.

There are two honest ways to state the result.

1. If large constants must be written literally, then the construction is an exact finite convex formulation but not a polynomial-bit-size one.

2. If one allows a polynomial-size spectrahedral-shadow gadget that constrains a variable to the singleton $\{2^{2^K}\}$, then the same construction is a polynomial-size succinct convex formulation. This is the interpretation under which the theorem gives a polynomial-size formulation.

The second interpretation uses nontrivial semidefinite-representation technology; it should not be replaced by the informal statement that repeated squaring alone enforces equality. Naive repeated-squaring inequalities generally give one-sided inequalities, whereas the singleton representation must force the value exactly.

This distinction is more than cosmetic. The theorem proves exact recoverability in a symbolic convex-optimization model. It does not prove that the formulation is numerically stable, practically useful, or solvable in strongly polynomial time. The choice of M deliberately creates an extremely sharp soft maximum, and this is precisely the sort of large-coefficient phenomenon that makes direct numerical solution unattractive.

The theorem also does not cover the following important cases without extra work:

- sa-rectangular ambiguity sets given by separation or membership oracles rather than by polynomially many listed vertices;
- polyhedral ambiguity sets with exponentially many vertices unless a compact exact vertex representation is supplied;
- ellipsoidal, Wasserstein, total-variation, KL, or general ϕ-divergence ambiguity balls, except in special cases where they reduce to polynomial finite vertex lists;
- s-rectangular ambiguity models coupling the actions within a state;
- undiscounted, average-reward, or stochastic-game variants not reducible to the discounted contraction setting above.

9 Conclusion

The argument verifies a genuine restricted result: finite, rational, vertex-listed sa-rectangular discounted RMDPs admit a one-shot lifted convex formulation from which the exact robust value vector can be recovered by rational reconstruction. This is partial progress on the convex-formulation program for robust MDPs, because it removes the limiting regularization step in this finite-vertex setting. It is not a full solution of the broader problem considered by Grand-Clement and Petrik, and it should not be advertised as such.

References

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